

9

Point groups

The theory of groups is a branch of mathematics in which one does something to something and then compares the results with the result of doing the same thing to something else, or something else to the same thing.

James Newman, mathematician (1956)

In this chapter, we will introduce the concept of a group, a set of symmetry operators that, together, describe the point symmetry of a 3-D crystalline object. We will see that there are precisely 32 such groups. This is an important result, because it allows us to classify any crystalline material as belonging to one of these 32 groups. We will introduce a number of mathematical tools that facilitate working with groups. As usual, the inclusion of magnetic symmetry (time-reversal) substantially complicates things, and we will explore how one obtains 122 magnetic or black-and-white point groups.

9.1

What is a group?

9.1.1

A simple example

Before using mathematical terminology to define what a group is, we will illustrate the concept using a practical example. Consider the crystal depicted in Fig. 9.1. This is a typical shape for a quartz crystal. Looking at the drawing, we find that the only symmetry elements present in this crystal are the ones corresponding to the configuration 223, discussed in the previous chapter. If we denote a general symmetry operator by the symbol $\mathcal{O}$, then we can define the following six symmetry operators for the quartz crystal:

$$\begin{aligned}\mathcal{O}_0 &= e = \text{the identity operator,} \\ \mathcal{O}_1 &= 3 = \text{rotation through an angle } \alpha = 2\pi/3, \\ \mathcal{O}_2 &= 3^2 = \text{rotation through an angle } \alpha = 4\pi/3, \\ \mathcal{O}_3 &= 2_x = \text{rotation through } \pi \text{ about the } x\text{-axis,} \\ \mathcal{O}_4 &= 2_y = \text{rotation through } \pi \text{ about the } y\text{-axis,} \\ \mathcal{O}_5 &= 2_u = \text{rotation through } \pi \text{ about the } u\text{-axis.}\end{aligned}\tag{9.1}$$

The rotation axes $2_{x,y,u}$ are perpendicular to the central axis 3. If 3 and e.g., 2_x are given, then, by Euler's theorem, the other two two-fold axes are automatically determined. We

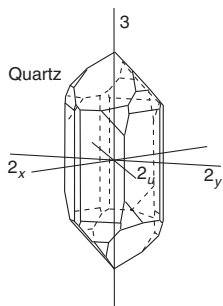


Figure 9.1 A typical quartz crystal and its symmetry elements.

define the successive operation of two symmetry operators by writing the operators in the reverse order, i.e., the operator that is applied first is written to the right:

$$\mathcal{O}_i.\mathcal{O}_j : \quad \mathcal{O}_j \text{ is applied first, followed by } \mathcal{O}_i. \quad (9.2)$$

The symbol $.$ stands for the multiplication or successive application of the two operators.

The successive operation of two symmetry operators is again a symmetry operator. Some examples are:

- $\mathcal{O}_1.\mathcal{O}_1 = \mathcal{O}_1^2 = \mathcal{O}_2$;
- $\mathcal{O}_1.\mathcal{O}_4 = \mathcal{O}_3$.

The inverse of every operator is also an operator:

- $\mathcal{O}_1$ and $\mathcal{O}_2$ are mutually reverse operators;
- $\mathcal{O}_3$, $\mathcal{O}_4$, and $\mathcal{O}_5$ are self-inverse operators.

And, finally, the identity operator e can also be obtained by combining the different operators, e.g., $e = \mathcal{O}_i.\mathcal{O}_i^{-1} = \mathcal{O}_1.\mathcal{O}_2 = \mathcal{O}_1^3 = \dots$

9.1.2

Group axioms

From the quartz example in the preceding section, we learn that the combination of two symmetry operators is again a symmetry operator, that there exists a *neutral* operator or *identity* operator, and that each operator has an inverse. In this section, we will formalize these statements and properly define what is a group.

The following four rules must be satisfied, in order for a set $\mathcal{G}$ to become a group under a certain multiplicative operation:

- (i) In the set $\mathcal{G}$, the multiplicative operation is defined such that the product of any pair of elements $\mathcal{O}_i, \mathcal{O}_j$ is also an element of $\mathcal{G}$, i.e., the set is *closed*:¹

$$\forall \mathcal{O}_i, \mathcal{O}_j \in \mathcal{G}, \exists \mathcal{O}_k \in \mathcal{G} : \mathcal{O}_i.\mathcal{O}_j = \mathcal{O}_k. \quad (9.3)$$

¹ $\forall$ means “for all, for each,” and $\exists$ means “there exists”; the colon “:” and $\rightarrow$ can be read as “such that” or “one has.” In words, Eq. (9.3) reads: for each pair of operators $\mathcal{O}_i, \mathcal{O}_j$ that belong to the set $\mathcal{G}$, there exists an operator $\mathcal{O}_k$, also in $\mathcal{G}$, such that the product of $\mathcal{O}_i$ and $\mathcal{O}_j$ equals $\mathcal{O}_k$.

9.1 What is a group?

(ii) The multiplicative operation is associative:

$$\mathcal{O}_i.(\mathcal{O}_j.\mathcal{O}_k) = (\mathcal{O}_i.\mathcal{O}_j).\mathcal{O}_k. \quad (9.4)$$

(iii) There exists a unit element e :

$$\exists e \in \mathcal{G} \rightarrow \forall \mathcal{O}_i \in \mathcal{G} : e.\mathcal{O}_i = \mathcal{O}_i = \mathcal{O}_i.e. \quad (9.5)$$

(iv) Each element has its inverse element:

$$\forall \mathcal{O}_i \in \mathcal{G}, \exists \mathcal{O}_k \in \mathcal{G} : \mathcal{O}_i.\mathcal{O}_k = \mathcal{O}_k.\mathcal{O}_i = e \quad ; \quad \mathcal{O}_k = \mathcal{O}_i^{-1}. \quad (9.6)$$

If a set of elements $\{\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_n\}$ satisfies rules (i) through (iv) for a given multiplication rule, then that set is called a group.

(v) The order of the operations is unimportant:

$$\forall \mathcal{O}_i, \mathcal{O}_j \in \mathcal{G} : \mathcal{O}_i.\mathcal{O}_j = \mathcal{O}_j.\mathcal{O}_i. \quad (9.7)$$

If, in addition, the set satisfies rule (v), then the group $\mathcal{G}$ is an Abelian or commutative group.

It is easy to show that the symmetry operators of the quartz example in the previous section satisfy all four principal rules; therefore, the set of symmetry operators of quartz forms a group. In general, the symmetry operators of any crystal form a group; as we will discuss in a later section, the crystallographic groups have definite designations (although there are different systems in use). The quartz group is named **32** (D_3); the first name is the internationally agreed upon name for this point group, i.e., the so-called *Hermann–Mauguin* symbol. The second name is the *Schoenflies* symbol, which is used primarily by solid state physicists and chemists.

The properties of any abstract group are fully determined if all possible multiplications between its elements are defined; for this purpose the concept of a *multiplication table* (or *Cayley table*) is introduced. The products of any two group-elements are written in a 2-D table; if every element appears only once in each row and column, then the set forms a group. If furthermore the square is symmetric with respect to the main diagonal, the group is *Abelian*. The multiplication order goes from the left-most column to the top row (i.e., elements of the top row are executed first).

As an example, we give here the full multiplication table of the quartz group:

32 (D_3)	e	3	3^2	2_x	2_y	2_u
e	e	3	3^2	2_x	2_y	2_u
3	3	3^2	e	2_u	2_x	2_y
3^2	3^2	e	3	2_y	2_u	2_x
2_x	2_x	2_y	2_u	e	3^2	3
2_y	2_y	2_u	2_x	3	e	3^2
2_u	2_u	2_x	2_y	3^2	3	e

9.1.3

Principal properties of groups

The following properties refer to all abstract groups.

- *Order of a group*: A group may contain one, several, or an infinite number of elements; this is called the *order* n of the group. If n is finite then the group is called a finite group.
Example: the order of the quartz point group is $n = 6$.
- *Isomorphism*: If a one-to-one correspondence exists between the elements of two groups, then those groups are isomorphous. Obviously, the order of isomorphous groups is the same. This relation between two groups is denoted by a double-headed arrow $\leftrightarrow$, i.e., if $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ and $\mathcal{H} = \{h_1, h_2, \dots, h_n\}$ then

$$\mathcal{G} \leftrightarrow \mathcal{H} \text{ if } \forall g_i, g_j \in \mathcal{G}, \forall h_i, h_j \in \mathcal{H} : g_i \cdot g_j \leftrightarrow h_i \cdot h_j. \quad (9.8)$$

Example: The group $\mathcal{G}$ denoting the space rotations through an angle $2\pi/N$ is isomorphous with the group of complex numbers $\exp(2\pi i n/N)$ where $0 \leq n < N$.

Isomorphous groups are, therefore, different representations of one and the same abstract group; consequently, isomorphous groups have identical multiplication tables.

- *Homomorphous groups*: Two groups are homomorphous if there exists a unidirectional correspondence between them, i.e., with every element of one group there corresponds one or more elements of the other group. This relation is denoted by a single-headed arrow $\rightarrow$, pointing from the group with the highest order to the one with the lowest order, i.e.,

$$\begin{aligned} \mathcal{G} \rightarrow \mathcal{H} \text{ if } \{g_{i_1}, \dots, q_{i_s}\} \rightarrow h_i, \{g_{j_1}, \dots, q_{j_t}\} \rightarrow h_j, \\ g_{i_p} \cdot g_{j_q} = h_i \cdot h_j \quad (1 \leq p \leq s, 1 \leq q \leq t). \end{aligned} \quad (9.9)$$

Mapping several elements of one group onto one element of another group can be useful in the classification of elements: one obvious example is the mapping of symmetry operators according to their handedness, i.e., a symmetry group $\mathcal{G}$ can be mapped onto the group $\{1, -1\}$. Another interpretation might be the following: if a symmetry operator changes the sign of e.g., the z -component of an atom coordinate vector, then this operator is mapped on -1 , else on $+1$. In the case of the quartz group **32** (D_3), this would produce the following mapping:

$$\mathbf{32} (D_3) \rightarrow \{1, -1\} : \{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\} \rightarrow 1, \{\mathcal{O}_3, \mathcal{O}_4, \mathcal{O}_5\} \rightarrow -1. \quad (9.10)$$

It is easy to verify that the set $\{1, -1\}$ is a group under ordinary multiplication.

- *Cyclic groups*: If a group contains an element $\mathcal{O}$, such that the powers of $\mathcal{O}$ exhaust all group elements, then this group is called a cyclic group, and $\mathcal{O}$ is called the *generating element*. The group can thus be written as:

$$\mathcal{G} = \{\mathcal{O}, \mathcal{O}^2, \dots, \mathcal{O}^l, \dots, \mathcal{O}^n = e\}. \quad (9.11)$$

This type of group represents rotations through an angle $2\pi/n$ and is denoted by C_n (Schoenflies notation) or **n** (Hermann–Mauguin symbol). For this type of group, it is sufficient to give the generating element instead of the full multiplication table.

9.2 3-D crystallographic point symmetries

- *Group generators*: Since all elements in a group can be written as products of the other elements (this is essentially shown in the group multiplication table) there will be a minimum number of elements from which all the others can be reconstructed. This minimal set of group elements is called the set of *group generators*. One can then reconstruct the complete group by (1) taking all the powers of each generator until either the identity or the generator itself is found, or (2) taking all the products between the resulting elements and the generators until no new elements are found.
- *Subgroups and supergroups*: Assume a group $\mathcal{G}$ of order n ; if a subset of n_k elements $\mathcal{O}_k$ also forms a group, then this set $\mathcal{G}_k$ is called a *subgroup* of $\mathcal{G}$ and is denoted as $\mathcal{G}_k \subset \mathcal{G}$. If, in addition, this subgroup is different from either $\{e\}$ or $\mathcal{G}$ itself, then the subgroup is called a *proper* subgroup.

The quartz group **32** (D_3) contains two proper subgroups:

$$\text{Subgroup } \mathbf{3} (C_3) = \{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}, \quad (9.12)$$

$$\text{Subgroup } \mathbf{2} (C_2) = \{\mathcal{O}_0, \mathcal{O}_3\}. \quad (9.13)$$

These groups describe pure three- and two-fold rotations, respectively, and are both cyclic.

Some groups do not have any proper subgroups; in that case only the *trivial* subgroups $\{e\}$ and $\mathcal{G}$ exist. The order of a subgroup is always an integer divider of the order of the main group (this is called the *theorem of Lagrange*); the ratio $p = n/n_k$ is called the *subgroup index*.

In a similar way, $\mathcal{G}$ is a *supergroup* of $\mathcal{G}_k$; a large part of group theory is devoted to the generation of complicated supergroups from simple, low-order groups.

9.2

3-D crystallographic point symmetries

In the preceding chapter, we derived all rotational symmetry operations compatible with the 14 Bravais lattices, and we have seen that the combination of symmetry elements generates new symmetry elements. These sets of symmetry elements, which are mutually consistent, are called *groups*. In this section, we will derive all 32 crystallographic *point groups*, i.e., all the symmetry groups that are compatible with translational periodicity in three dimensions. They are known as point groups because all symmetry elements intersect each other in a single point.

The 32 crystallographic point groups can be derived in a very simple way by counting the ways in which the symmetry elements can be combined to form closed sets. To derive all groups in an unambiguous way, we will proceed in seven steps, from the simplest groups to the highest-order cubic groups. In the following, all point groups will be denoted with both the international (or Hermann–Mauguin) symbol and the Schoenflies symbol (in brackets). Note that all point groups will be discussed, and the crystallographic ones will be highlighted.

To clarify the meaning of the various combinations of symmetry elements, rendered 3-D illustrations are included in the following sections; these images were created with

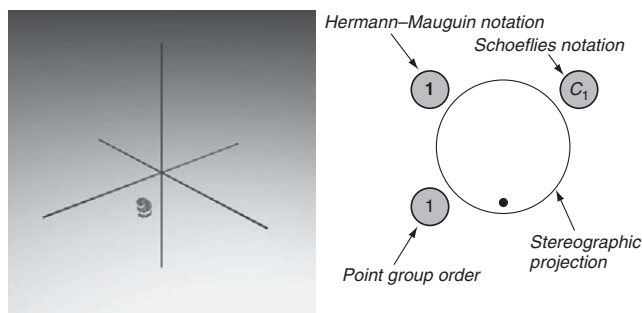


Figure 9.2 Graphic representation of the identity point group $1 (C_1)$.

a ray-tracing program (*Rayshade*, De Graef, 1998),² and show the symmetry elements in the proper orientation in space, along with sets of equivalent points. These points are organized as a short helix, to highlight the fact that some of the symmetry elements change the handedness of the object, whereas others leave it unchanged. For each point group, the corresponding stereographic projection is also shown. Note that the handedness of the equivalent points is not usually shown in these stereographic projections; they only distinguish between points in the northern and southern projection hemispheres.

The simplest point group contains only the identity operator $1 (E)$; its schematic representation is shown in Fig. 9.2. The corresponding stereographic projection is shown to the right of the figure: three labels are shown, one for the Hermann–Mauguin symbol (top left), one for the Schoenflies notation (top right), and one for the order of the point group (lower left).

Before we proceed with the derivation, it is useful to make a few comments on the point group notation, in particular for the international or Hermann–Mauguin notation. In general, the Hermann–Mauguin notation consists of (at most) three symbols; each symbol corresponds to a particular direction in the Bravais lattice. The relevant directions are listed in Table 9.1. When we name the crystallographic point groups in the following subsections, we will refer to this table.

9.2.1

Step I: the proper rotations

The simplest groups are the proper rotation groups, formed by the identity and all powers of an n -fold rotation axis, which is the generating element.

Notation: $n [C_n]$ [C for cyclic].

All rotational groups are obviously cyclic groups of order n ; they contain a single invariant line (the rotation axis itself). All odd groups are polar in all directions, whereas the

² Rayshade 4.0 ray-tracing system (public domain software). www-graphics.stanford.edu/~cek/rayshade/rayshade.html.

9.2 3-D crystallographic point symmetries

Table 9.1 Primary, secondary, and tertiary symmetry directions in each of the seven crystal systems. For the tetragonal system, the symbol $\langle uvw \rangle$ refers to the fact that equivalent directions are obtained from permutations of the first two indices only.

Crystal system	Primary $[uvw]$	Secondary $[uvw]$	Tertiary $[uvw]$
Cubic	$\langle 100 \rangle$	$\langle 111 \rangle$	$\langle 110 \rangle$
Hexagonal	$[00.1]$	$[10.0]$	$[12.0]$
Tetragonal	$[001]$	$\langle 100 \rangle$	$\langle 110 \rangle$
Orthorhombic	$[100]$	$[010]$	$[001]$
Trigonal	$[111]$	$[010]$	$[1\bar{1}0]$
Monoclinic	$[010]$	—	—
Triclinic	—	—	—

even groups are only polar along the rotation axis.³ If $N = nm$ then both $\mathbf{n}$ and $\mathbf{m}$ are subgroups of $\mathbf{N}$; as an example, consider the group $\mathbf{6}$ (C_6) of order 6, which has both $\mathbf{2}$ (C_2) and $\mathbf{3}$ (C_3) as subgroups. For $n \rightarrow \infty$, the rotation group describes the symmetry of a rotating cone. The crystallographic point (rotation) groups are $\mathbf{1}$ (C_1), $\mathbf{2}$ (C_2), $\mathbf{3}$ (C_3), $\mathbf{4}$ (C_4), and $\mathbf{6}$ (C_6) and they are shown graphically in Figs. 9.2 and 9.3. Using the symmetry directions of Table 9.1, we find for the monoclinic system, for which the two-fold rotation is the highest-order rotational symmetry, that the rotation axis is parallel to the primary direction, which is the $[010]$ direction. For the trigonal system, on the other hand, the primary direction is the $[111]$ direction, so that the three-fold axis in this system is parallel to $[111]$. In the hexagonal system, the six-fold axis lies along the primary $[00.1]$ direction, and in the tetragonal system, the four-fold rotation axis lies along $[001]$.

9.2.2

Step II: combining proper rotations with two-fold rotations

According to Euler's theorem, the presence of one two-fold axis perpendicular to a rotation axis will result in the generation of a set of equivalent two-fold axes. This generates the so-called *dihedral* point groups.

Notation: $\mathbf{n2}$ [D_n] [D for dihedral].

Once again, there is a difference between the odd and the even groups: for the odd groups the operator n generates all two-fold axes, as shown in Fig. 9.4 for the group $\mathbf{32}$ (D_3). For all these groups the n -fold axis is non-polar, whereas the two-fold axes are polar. The group $\mathbf{12=2}$ (C_2) is already contained in the cyclic rotation groups.

³ A group is polar if the directions $\mathbf{t}$ and $-\mathbf{t}$ are not related to each other by a symmetry operation.

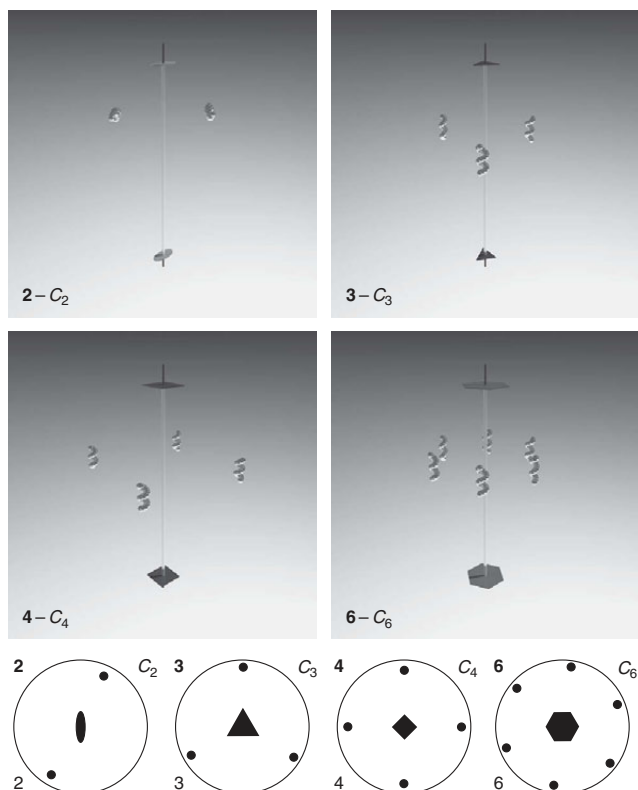


Figure 9.3 Graphic representation of the crystallographic cyclic point groups.

For the even groups on the other hand, not all two-fold axes are generated from just one initial axis; a second two-fold axis needs to be provided in order to generate the complete set. Therefore, the Hermann–Mauguin symbol contains two **2** (C_2)s. The even dihedral groups are all non-polar.

The order of all dihedral groups is $2n$. The crystallographic dihedral point groups are **222** (D_2), **32** (D_3), **422** (D_4), and **622** (D_6) (see Fig. 9.4).

In the trigonal system, the three-fold axis lies along the $[111]$ direction, as before, and the two-fold axis lies along the tertiary ($\bar{1}10$) directions, since these directions are at right angles to $[111]$. In the tetragonal and hexagonal systems, the two-fold axes lie along both secondary and tertiary directions (see Table 9.1), whereas the orthorhombic system has a two-fold axis along each of the three symmetry directions.

When the order of the point group increases to infinity, we obtain the group $\infty 2$; this group represents the symmetry of a cylinder in which the top surface rotates in the opposite direction of the bottom surface, as illustrated in Table 9.3 on page 215.

9.2 3-D crystallographic point symmetries

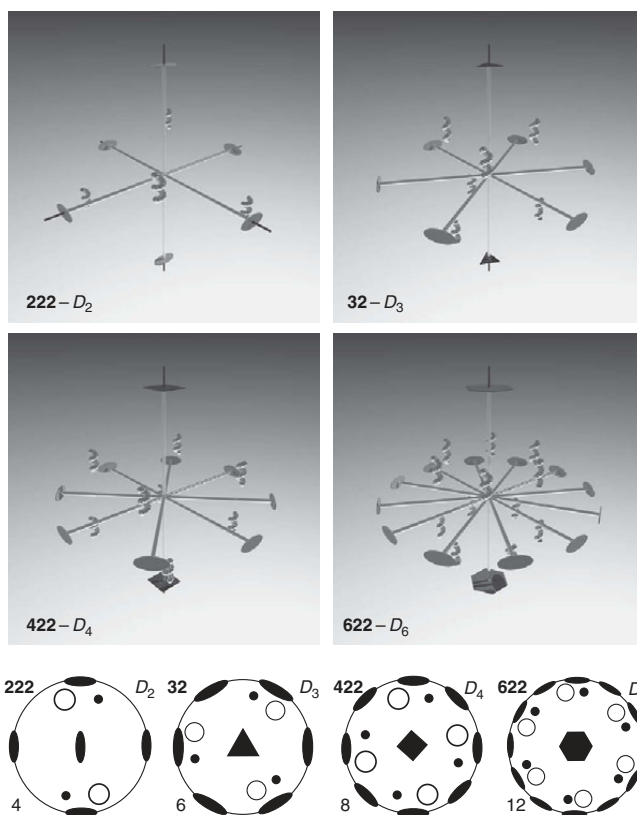


Figure 9.4 Graphic representation of the crystallographic dihedral point groups.

9.2.3

Step IIIa: combining proper rotations with inversion symmetry

These groups are obtained by combination of an n -fold rotation axis with a center of symmetry in the origin. Each inversion rotation group has equivalence with a mirror-rotation group $\tilde{n}$: $\tilde{n}_{\text{odd}} \leftrightarrow 2n$ and $\tilde{n}_{\text{odd}} \leftrightarrow 2n$.

Notation: $\bar{n} [S_n]$.

Note that these groups have one single axis but contain operators of the second kind. This is particularly clear in the 3-D renderings, which show the presence of both left-handed and right-handed helices.

For $n = 4k$, the inversion groups and mirror-rotation groups are identical, e.g., $\bar{4} = \tilde{4}$, $\bar{8} = \tilde{8}$, ... For $n = 4k + 2$, the inversion element actually generates a simple mirror plane

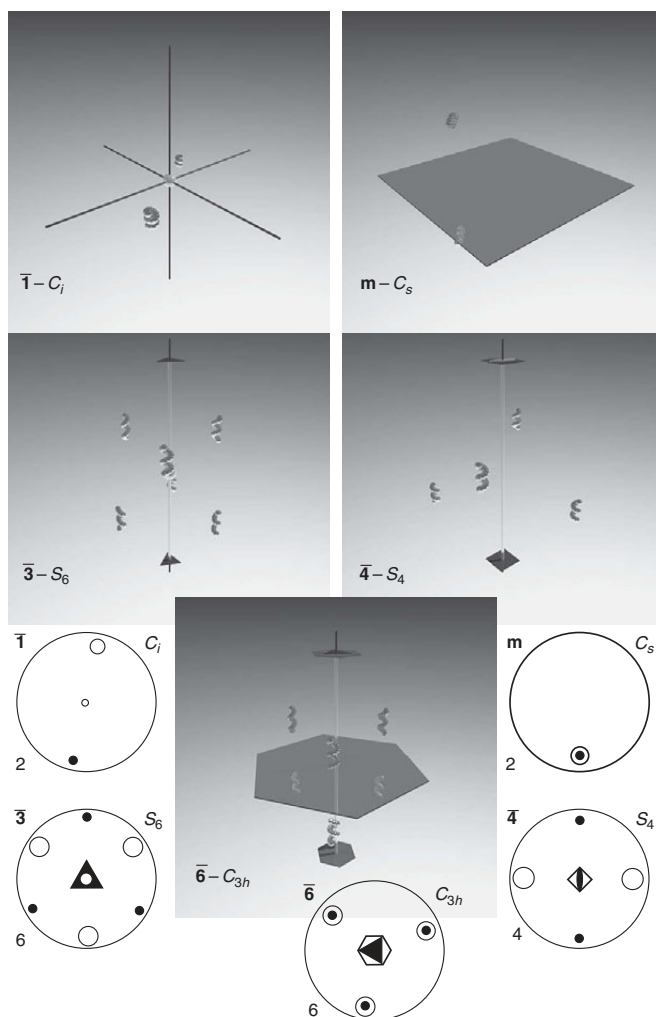


Figure 9.5 Graphic representation of the crystallographic $\bar{n}$ point groups.

perpendicular to the rotation axis; these groups are the odd groups of step IIIb (see next section). Normally, $\bar{6} (C_{3h})$ is written as $3/m$.

The crystallographic inversion rotation groups are $\bar{1} (C_i)$, $\bar{2} (=m (C_s))$, $\bar{3} (C_{3i})$, $\bar{4} (S_4)$, and $\bar{6} (C_{3h})$ ($=3/m$) and they are shown schematically in Fig. 9.5. The order of these point groups is $2n$ when n is odd, and n when n is even. The direction of the roto-inversion axes

9.2 3-D crystallographic point symmetries

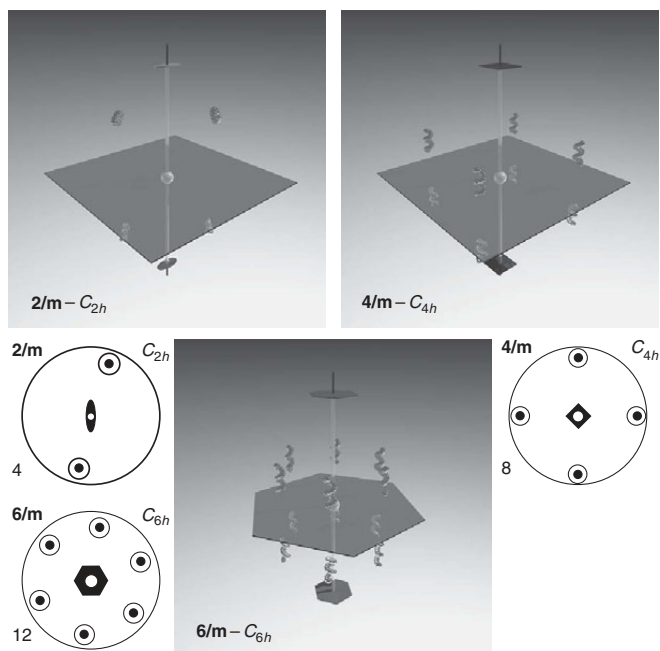


Figure 9.6 Graphic representation of the crystallographic n/m point groups.

in the various crystal systems is the same as that of the proper rotation axes discussed in Section 9.2.1.

9.2.4

Step IIIb: combining proper rotations with perpendicular reflection elements

These groups contain a principal axis and a mirror plane perpendicular to the axis; hence, they contain operations of the second kind and both left-handed and right-handed helices in the 3-D representations.

Notation: $n/m [C_{nh}]$ [h for horizontal]

The odd members of this type of point group were already discussed in the previous section; the point groups n_{odd}/m are equivalent to $\bar{2}n$. The order of all these groups is $2n$. The limiting group of this type describes a rotating cylinder ∞/m , with top and bottom faces rotating in the same sense.

The crystallographic point groups of type n/m are $2/m$ (C_{2h}), $4/m$ (C_{4h}) and $6/m$ (C_{6h}) (see Fig. 9.6). The direction of the rotation axis is given by the primary direction for each crystal system in Table 9.1, and the Miller indices of the mirror plane correspond to

the plane perpendicular to the rotation axis in each crystal system. For instance, for the monoclinic point group $2/\mathbf{m}$ (C_{2h}), the two-fold axis lies along the $[010]$ direction and the (010) plane is the mirror plane.

9.2.5

Step IV: combining proper rotations with coinciding reflection elements

Mirror planes containing a rotation axis of order n generate a new type of point group; the generating elements are $\mathbf{n}$ and $\mathbf{m}$.

Notation: $\mathbf{nm}$ [C_{nv}] [v for vertical].

The situation here is very similar to the one in Step II: odd rotation axes generate all the mirror planes from a single starting plane, whereas for even rotation axes a second generating mirror plane must be provided, as illustrated in Fig. 9.7. The order of all these groups is $2n$. The group $1\mathbf{m}$ is identical to $\mathbf{m}$ (C_s) = $\bar{2}$. The limiting symmetry $\infty\mathbf{mm}$ describes a regular cone.

The crystallographic point groups of type $\mathbf{nm}$ are $\mathbf{mm2}$ (C_{2v}), $3\mathbf{m}$ (C_{3v}), $4\mathbf{mm}$ (C_{4v}), and $6\mathbf{mm}$ (C_{6v}). Note that the two-fold axis in the point group $\mathbf{mm2}$ (C_{2v}) is written as the third symbol, not the first one. This is in agreement with the entries for the orthorhombic crystal system in Table 9.1, which show that the $[001]$ -axis (parallel to the two-fold axis) is the tertiary direction. For all other crystal systems, the rotation axis falls along the primary direction for this class of point groups.

9.2.6

Step Va: combining inversion rotations with coinciding reflection elements

The combination of an inversion point and a reflection plane through this point is equivalent to a two-fold axis. The generators are the inversion rotation and the mirror plane.

Notation: $\bar{\mathbf{n}}\mathbf{m}$ [D_{nd}] [d for diagonal].

In odd and “twice-even” ($n = 4k$) groups, the two-fold axes bisect the angles between the mirror planes. For odd groups only one mirror plane is needed in the international symbol, for even groups a two-fold axis and a mirror plane are considered to be the generators. The simplest groups $\bar{1}\mathbf{m}$ and $\bar{2}\mathbf{m}$ are equivalent to $2/\mathbf{m}$ (C_{2h}) and $\mathbf{mm2}$ (C_{2v}), respectively.

The crystallographic point groups of type $\bar{\mathbf{n}}\mathbf{m}$ are $\bar{3}\mathbf{m}$ (D_{3d}), $\bar{4}2\mathbf{m}$ (D_{2d}), and $\bar{6}2\mathbf{m}$ (D_{3h}) and are shown in Fig. 9.8. In $\bar{4}2\mathbf{m}$ (D_{2d}), the two-fold axes are considered to lie along the secondary directions of the type $\langle 100 \rangle$, whereas the mirror planes lie perpendicular to the tertiary directions (see Table 9.1). For $\bar{6}2\mathbf{m}$ (D_{3h}) on the other hand, the mirror planes are taken to be normal to the secondary directions $\langle 10.0 \rangle$ and the two-fold axes lie along the tertiary directions, hence the order of the symbols $\mathbf{m}$ and 2 is reversed in the two point group symbols. Note that for $\bar{6}2\mathbf{m}$ (D_{3h}), the two-fold axes lie in the mirror planes, whereas in $\bar{4}2\mathbf{m}$ (D_{2d}), the mirror planes bisect the two-fold axes.

9.2 3-D crystallographic point symmetries

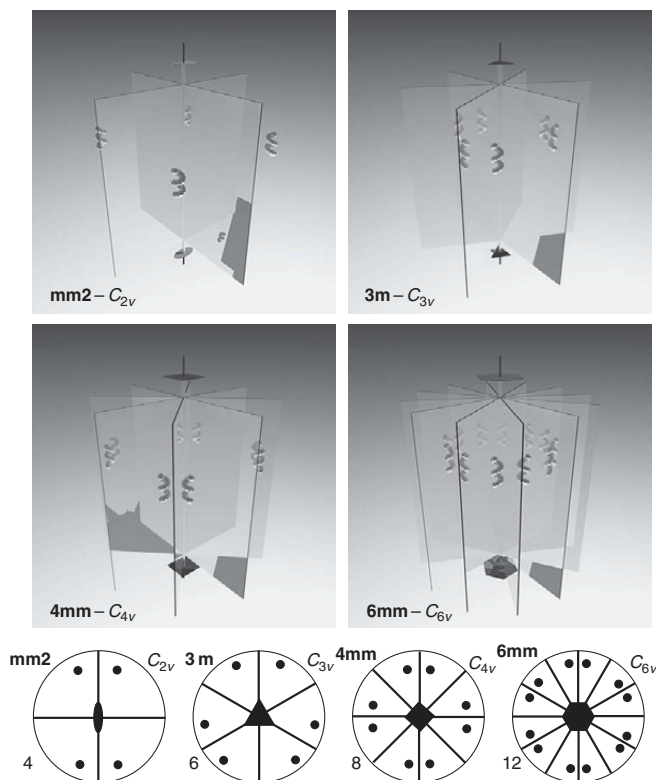


Figure 9.7 Graphic representation of the crystallographic nm point groups.

9.2.7

Step Vb: combining proper rotations with coinciding and perpendicular reflection elements

If both perpendicular and coinciding mirror planes are present, then only the even rotations create new groups; the rotation axis and one mirror plane of each type are needed as generators.

Notation: $\frac{n}{m}m [D_{nh}]$.

The intersection of two mirror planes at right angles generates a two-fold axis along the intersection line. Again, two independent coinciding mirror planes need to be defined for the even rotation groups. The full symbols of the groups are $\frac{2}{m}\frac{2}{m}\frac{2}{m}$, $\frac{4}{m}\frac{2}{m}\frac{2}{m}$, and $\frac{6}{m}\frac{2}{m}\frac{2}{m}$. The order of all these groups is $4n$.

The crystallographic point groups of type $\frac{2}{m}m$ are $mmm (D_{2h})$, $4/mmm (D_{4h})$ and $6/mmm (D_{6h})$ (in shorthand notation) and the graphical representations are shown in

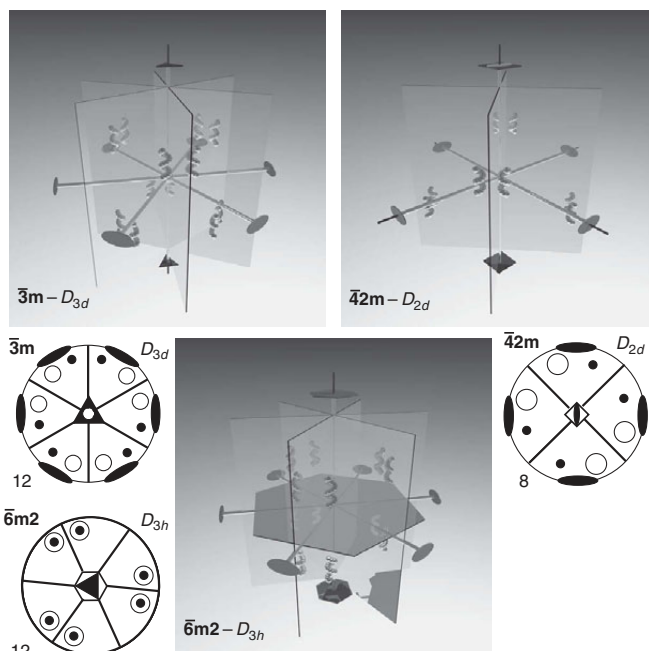


Figure 9.8 Graphic representation of the crystallographic $\bar{n}m$ point groups.

Fig. 9.9. The rotation axes are, as before, oriented along the primary directions of the crystal systems, and the mirror planes are perpendicular to both the secondary and tertiary directions.

9.2.8

Step VI: combining proper rotations

The only combinations of rotational symmetries we have not used yet are 233 and 432. These generate groups with only rotational elements present.

Notation: $n_1n_2 [T]$ and $[O]$.

There are only three point groups of this type, namely **23** (*T*), **432** (*O*), and the (non-crystallographic) icosahedral point group **532** (*I*). The Schoenflies symbols for the crystallographic point groups stand for tetrahedral and octahedral symmetry, respectively. The orders of these point groups are 12, 24, and 60, respectively. Note that they are groups with only operators of the first kind. Because of the presence of multiple three-fold axes, the point groups **23** (*T*) and **432** (*O*) belong to the cubic symmetry type. All

9.2 3-D crystallographic point symmetries

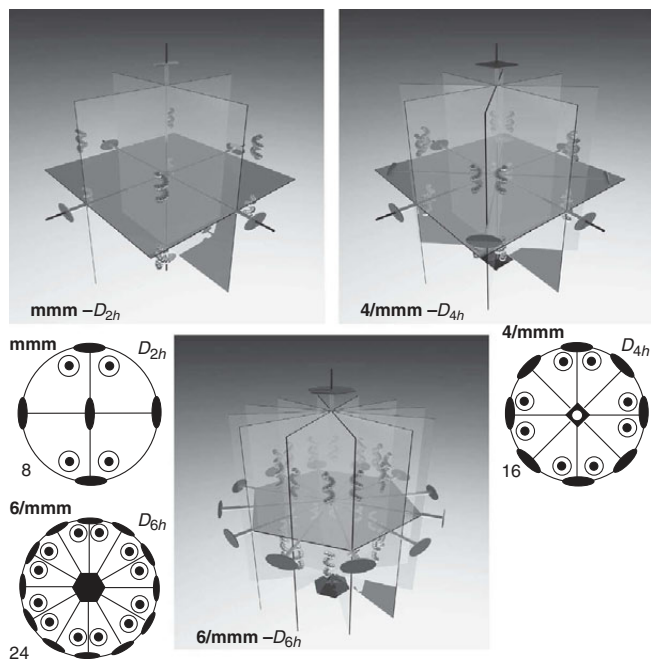


Figure 9.9 Graphic representation of the crystallographic $\frac{n}{m}m$ point groups.

equivalent points have the same handedness, since there are no symmetry operators of the second kind.

The crystallographic point groups of type n_1n_2 are **23** (T) and **432** (O), and schematic drawings are shown in Fig. 9.10. In point group **23** (T), the two-fold axis is oriented along the primary direction of the cubic crystal system (see Table 9.1), with the three-fold axes along the secondary directions. Note that there is also a point group **32** (D_3), so the order of the symbols is important! In point group **432** (O), the four-fold axes lie along the primary directions, the three-fold axes, as before, along the secondary directions, and the two-fold axes lie along the tertiary directions of the cubic crystal system.

9.2.9

Step VII: adding reflection elements to Step VI

Finally, we can add mirror planes (or inversion symmetry) to the groups of Step VI; because of the mutual orientation of the rotation axes, these symmetry planes will in general be arranged in an “oblique” way.

Notation: $\bar{n}_1n_2$ [T_h], [T_d], and [O_h].

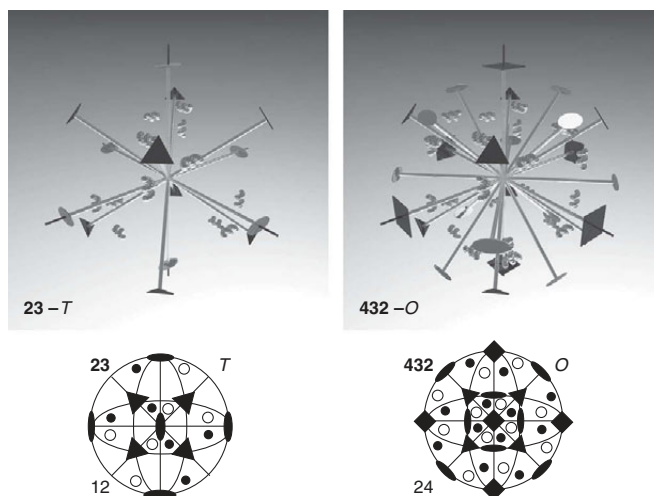


Figure 9.10 Graphic representation of the crystallographic $\bar{n}_1 n_2$ point groups.

If we add mirror planes through the two-fold axes of $23 (T)$, the three-fold axes will become inversion axes and the resulting point group is of order 24 and is denoted by the symbol $\bar{m}\bar{3} (T_h)$.

Adding mirror planes through the three-fold axes of $23 (T)$ results in four-fold inversion axes (but note that the inversion element itself is not present in this group!); the resulting group is of order 24 and denoted by $\bar{4}3m (T_d)$.

Finally, we can add either mirror planes or the inversion symmetry to the group $432 (O)$ to obtain the crystallographic group of the highest-order (48) which is denoted by $\bar{m}\bar{3}m (O_h)$. In full notation this group is known as $\frac{4}{m}\frac{3}{m}\frac{2}{m}$. Combination of mirror planes with $532 (I)$ results in the icosahedral group $\bar{m}\bar{3}5 (I_h)$ of order 120.⁴

The crystallographic point groups of type $\bar{n}_1 n_2$ are $\bar{m}\bar{3} (T_h)$, $\bar{4}3m (T_d)$, and $\bar{m}\bar{3}m (O_h)$ and they are shown in Fig. 9.11. The orientation of the rotation axes is identical to that of the point groups $23 (T)$ and $432 (O)$. This concludes the enumeration of all 32 crystallographic point groups.

9.2.10

General remarks

9.2.10.1 Classes of point groups Out of the 32 crystallographic point groups, only 11 have a center of symmetry. These *centrosymmetric* point groups are shown in the

⁴ The icosahedral symmetry groups will be discussed in much more detail in the chapter on non-crystallographic symmetry (Chapter 15).

9.2 3-D crystallographic point symmetries

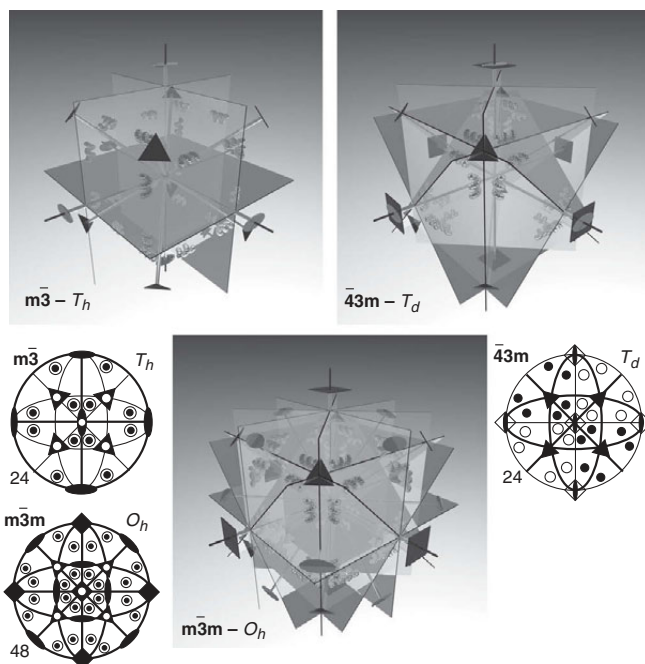


Figure 9.11 Graphic representation of the crystallographic $\bar{n}_1 n_2$ point groups.

second column of Table 9.2. The other non-centrosymmetric point groups are subgroups of these 11 groups. It is not difficult to verify that each of the groups in the last column of Table 9.2 becomes equivalent to the centrosymmetric one when combined with the inversion operator. All the groups in a single row of the table belong to a so-called *Laue class*.⁵ The centrosymmetric point group in a Laue class is used as the class symbol.

Polar point groups are groups in which there exists at least one direction that has no symmetrically equivalent directions. It is easy to see that this can only happen in non-centrosymmetric point groups in which there is at the most a single rotation axis. There are ten such polar groups: they are **4mm** (C_{4v}) and **6mm** (C_{6v}) and all their subgroups **6** (C_6), **4** (C_4), **3m** (C_{3v}), **3** (C_3), **mm2** (C_{2v}), **m** (C_s), **2** (C_2), and **1** (C_1).

The presence of symmetry at the atomic length scale has far-reaching consequences for the properties of a given material, whether they be mechanical, optical, electrical, magnetic, or thermal. It can be shown in very general terms that the symmetry group of *any* property of a crystal must include the point group symmetry operations of that

⁵ The reason for this terminology will become clear when we discuss diffraction in Chapter 11.

Table 9.2 The seven crystal systems with the centrosymmetric point groups (Laue classes), and the remaining point groups that are subgroups of the 11 centrosymmetric groups.

Crystal system	Laue class	Lower-symmetry class members
Triclinic	$\bar{1}$ (C_i)	1 (C_1)
Monoclinic	$2/m$ (C_{2h})	2 (C_2), m (C_s)
Orthorhombic	mmm (D_{2h})	222 (D_2), $mm2$ (C_{2v})
Tetragonal	$4/m$ (C_{4h})	4 (C_4), $\bar{4}$ (S_4)
	$4/mmm$ (D_{4h})	422 (D_4), $4mm$ (C_{4v}), $\bar{4}2m$ (D_{2d})
Trigonal	$\bar{3}$ (C_{3i})	3 (C_3)
	$\bar{3}m$ (D_{3d})	32 (D_3), $3m$ (C_{3v})
Hexagonal	$6/m$ (C_{6h})	6 (C_6), $\bar{6}$ (C_{3h})
	$6/mmm$ (D_{6h})	622 (D_6), $6mm$ (C_{6v}), $\bar{6}m2$ (D_{3h})
Cubic	$m\bar{3}$ (T_h)	23 (T)
	$m\bar{3}m$ (O_h)	432 (O), $\bar{4}3m$ (T_d)

crystal. This is known as *Neumann's principle* (Newnham, 2004). Certain material properties, such as piezoelectricity and optical activity, can occur only in non-centrosymmetric point groups. The application of symmetry to material properties is rich and extensive but outside the scope of this text; we refer the interested reader to the literature (Newnham, 2004, Nye, 1957).

9.2.10.2 Chirality and enantiomorphism The stereographic projection is the standard method of representing the point groups graphically. The rendered images shown in Figs. 9.2 through 9.11 have the added advantage that the handedness of each equivalent point is easily observed. If all equivalent points have the same handedness, then the object with the corresponding point group can exist in two different versions: a left-handed version and a right-handed version. The point groups for which this can occur have no improper rotations, i.e., no roto-inversions or mirror-rotations. Crystals or molecules that belong to these point groups are known as *chiral objects*, i.e., they have a handedness. Such objects are also known as *dissymmetric objects* (Hahn, 1996, p. 787). In chemistry and biology, the terms *enantiomerism* and *chirality* are used, whereas in crystallography the term *enantiomorphism* is more common.

9.2.10.3 Matrix representation of point groups In the previous chapter, we have seen that every symmetry operation can be represented as a matrix. Since the point group operations do not contain any translations, we can restrict ourselves to the 3×3 sub-matrix $\mathbf{D}$ of the full Seitz symbol $(\mathbf{D}|\mathbf{t})$. Returning to the quartz point group 32 (D_3) introduced in the first section of this chapter, we can associate with each symmetry operation $\mathcal{O}_i$ a 3×3 matrix $\mathbf{D}_i$. If we take the axes 2_x , 2_y , and 3 of Fig. 9.1 as the reference axes, then it is easy to show that the six transformation matrices are given by:

9.2 3-D crystallographic point symmetries

$$\begin{aligned}
D_0 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & D_1 &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\
D_2 &= \begin{pmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & D_3 &= \begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \\
D_4 &= \begin{pmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, & D_5 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.
\end{aligned} \tag{9.14}$$

Using these matrices, we can construct a multiplication table, similar to the one on page 195, by taking all possible products between two matrices:

$32 (D_3)$	D_0	D_1	D_2	D_3	D_4	D_5
D_0	D_0	D_1	D_2	D_3	D_4	D_5
D_1	D_1	D_2	D_0	D_5	D_3	D_4
D_2	D_2	D_0	D_1	D_4	D_5	D_3
D_3	D_3	D_4	D_5	D_0	D_2	D_1
D_4	D_4	D_5	D_3	D_1	D_0	D_2
D_5	D_5	D_3	D_4	D_2	D_1	D_0

Note that these two tables have exactly the same structure! This is an example of an *isomorphism* between the group of the operators $\mathcal{O}_i$ and the group of the six matrices shown above. Both of these are *representations* of an underlying abstract group, $\mathcal{G}$, with elements $\{a, b, c, d, e, f\}$, which can be defined by its own multiplication table; the same can be done for all other point groups as well. It is possible for multiple point groups to be isomorphic with a single abstract group.

$\mathcal{G}$	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	a	f	d	e
c	c	a	b	e	f	d
d	d	e	f	a	c	b
e	e	f	d	b	a	c
f	f	d	e	c	b	a

9.2.10.4 Group-subgroup relations; descent in symmetry Figure 9.12 shows all the group-subgroup relationships between the crystallographic point groups. The two highest-order point groups are the hexagonal group $6/\text{mmm}$ (D_{6h}) and the cubic group $\text{m}\bar{3}\text{m}$ (O_h);

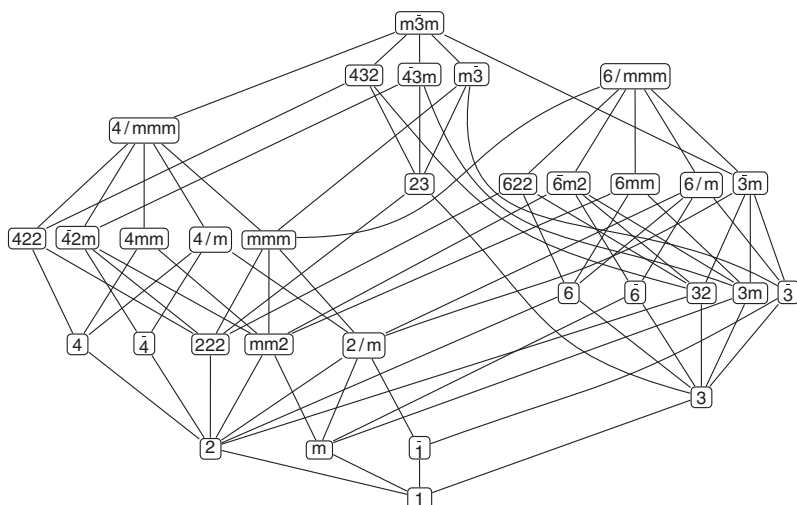


Figure 9.12 Descent in symmetry for the crystallographic point groups, illustrating group–subgroup relationships and the point group orders (left column). Only the international point group symbols are shown.

all other groups can be considered as subgroups of these two. In some cases, there are several possible orientations of a subgroup with respect to the larger group. For instance, point group **3** (C_3), a simple three-fold rotation axis, has four possible orientations with respect to point group **23** (T). On the other hand, **222** (D_2) has the same orientation as **23** (T), as is readily seen from a comparison of the two stereographic projections. The chart can also be used to determine all the centrosymmetric point groups; simply start from **1** (C_i) and list all the point groups that **1** (C_i) is a subgroup of (i.e., follow all the lines originating in **1** (C_i)). The centrosymmetric point groups are: **1** (C_i), **2/m** (C_{2h}), **4/m** (C_{4h}), **mmm** (D_{2h}), **4/mmm** (D_{4h}), **m3m** (O_h), **3** (C_{3i}), **m3** (T_h), **6/m** (C_{6h}), **3m** (D_{3d}), and **6/mmm** (D_{6h}), in agreement with the 11 Laue classes listed in Table 9.2. Figure 9.12 is known as a *descent in symmetry*, and we will use this representation again when we discuss the non-crystallographic point groups in Chapter 15.

9.2.10.5 Special positions and orbits All the point group illustrations in the preceding sections use as their general object a right-handed helix. Care has been taken to make sure that this helix does not intersect any of the symmetry elements of the group. When the object *is* allowed to intersect one or more of the symmetry elements, then something interesting happens. Figure 9.13(a) shows a rendered representation of point group **4/mmm** (D_{4h}). The helix has been replaced by a small sphere. When the sphere does not intersect any of the symmetry elements, there are a total of 16 equivalent spheres, which is the order of the point group. These positions are known as *general positions*. We define the *site symmetry* of a particular location as the point group of the object

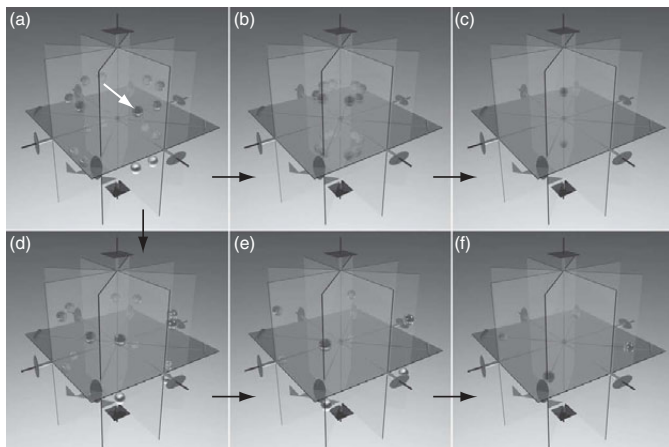


Figure 9.13 Rendered representation of point group $4/mmm$ (D_{4h}) and a general point orbit (a). In (b), the point moves towards the z -axis, and in (c) the special position $(0, 0, z)$ is shown. In (d), the general point moves towards one of the mirror planes and results in special point (x, x, z) in (e). In (f), the special point of the type $(x, x, 0)$ is shown.

with respect to that location. For the sphere arrowed in Fig. 9.13(a), there are no symmetry elements going through the center of the sphere, so the site symmetry for that sphere is equal to 1 (C_1). This is always the case for a general position, and, in fact, we can define the general position for any point group symmetry as a position with site symmetry 1 (C_1).

The set of all points that are symmetrically equivalent to a point with coordinates (x, y, z) with respect to a particular point group $\mathcal{G}$ is called the *crystallographic orbit* (or, simply, the *orbit*) of (x, y, z) with respect to $\mathcal{G}$. The number of points in the orbit of a general point with respect to a point group is equal to the order of that point group. For the example in Fig. 9.13(a), the orbit consists of the following points:

$$\begin{array}{cccc}
 (x, y, z) & (x, -y, z) & (-x, y, z) & (-x, -y, z) \\
 (x, y, -z) & (x, -y, -z) & (-x, y, -z) & (-x, -y, -z) \\
 (y, x, z) & (-y, x, z) & (y, -x, z) & (-y, -x, z) \\
 (y, x, -z) & (-y, x, -z) & (y, -x, -z) & (-y, -x, -z)
 \end{array} \quad (9.15)$$

These coordinates can be obtained by operating with all the symmetry matrices of the point group $4/mmm$ (D_{4h}) on the point with general coordinates (x, y, z) . The general position used for Fig. 9.13(a) is equal to $(0.65, 0.15, 0.40)$.

When we move the general point towards the z -axis, as shown in Fig. 9.13(b) and (c), then the 16 equivalent points will “merge” into only two points located on the z -axis. This means that for the point $(0, 0, z)$, the orbit consists of only two points, $(0, 0, z)$ and $(0, 0, -z)$. This is also obvious from Eq. 9.15: when we put $x = 0$ and $y = 0$, there are only two distinct points left over. Note that the point $(0, 0, z)$ is located on the four-fold

rotation axis, and also on the four mirror planes that contain the four-fold axis, but not on the horizontal mirror plane. Hence, the site symmetry of $(0, 0, z)$ is **4mm** (C_{4v}). Points like this are known as *special points*, and their site symmetry is higher than that of the general position with point group **1** (C_1).

Alternatively, we can move the point (x, y, z) to one of the vertical mirror planes, as shown in Fig. 9.13(d) and (e). On the mirror plane, the coordinate of the point is (x, x, z) , and it is clear that there are only eight equivalent points. The site symmetry is simply a single mirror plane, represented by the point group **m** (C_s). However, we can say something more about this mirror plane, namely that it is oriented at 45° with respect to the coordinate axes. Referring to Table 9.1, we see that the $\langle 110 \rangle$ directions in the tetragonal crystal system represent the tertiary symmetry directions. We include this information in the Hermann–Mauguin symbol for the site symmetry as follows: $\cdot \cdot \mathbf{m}$. The two dots indicate that there are no symmetry elements along the primary and secondary directions. If we had moved the general point to the $(x, 0, z)$ position, then there would again be eight equivalent positions, but this time the site symmetry would be $\cdot \mathbf{m} \cdot$, since the mirror plane is normal to the secondary symmetry direction for the tetragonal system.

Finally, we move the special point (x, x, z) down to the horizontal mirror plane in Fig. 9.13(f). The result is an orbit with four equivalent points. The site symmetry consists of two intersecting mirror planes (at right angles). We know that this corresponds to point group **mm2** (C_{2v}), but once again we can include a little additional information in the site symmetry symbol. The horizontal mirror plane is normal to the primary tetragonal symmetry direction. Both the vertical mirror plane and the two-fold axis along the intersection of the two mirror planes are located along the tertiary direction, so that the site symmetry is written as $\mathbf{m} \cdot \mathbf{m2}$. If we had moved the special point to the $(x, 0, 0)$ position, then the orbit would again contain four points, this time with site symmetry **m2m**.

Let us now summarize the general and special positions that we have found for the point group **4/mmm** (D_{4h}). We start with the orbit with the largest number of members, which is the general position (x, y, z) , and then list the special positions in descending order:

16	<i>g</i>	1	(x, y, z)	(hkl)	
8	<i>f</i>	$\cdot \mathbf{m} \cdot$	$(x, 0, z)$	$(h0l)$	
8	<i>e</i>	$\cdot \cdot \mathbf{m}$	(x, x, z)	(hhl)	
8	<i>d</i>	$\mathbf{m} \cdot \cdot$	$(x, y, 0)$	$(hk0)$	
4	<i>c</i>	m2m	$(x, 0, 0)$	(100)	(9.16)
4	<i>b</i>	$\mathbf{m} \cdot \mathbf{m2}$	$(x, x, 0)$	(110)	
2	<i>a</i>	4mm	$(0, 0, z)$	(001)	.

The last column contains Miller indices; for the (hhl) -type planes, there are eight equivalent planes, just as there are eight equivalent positions for the $(x, 0, z)$ point. The first column is then known as the *multiplicity* of the plane in the point group **4/mmm** (D_{4h}).

The letter symbols in the second column are added once all the special positions are listed. By convention, the highest-order site symmetry is labeled by the letter *a*, the next one by *b*, and so on, until all site symmetries have been labeled. If we combine the first two columns, then we obtain what is known as a *Wyckoff position*. For instance, we can talk about the *8e* position, or the *4b* position. Chapter 10 in the *International Tables for Crystallography, Volume A*, standardizes all possible site symmetries for all crystallographic

Table 9.3 Crystallographic and non-crystallographic point groups.

I n	II n2	IIIa $\bar{n}$	IIIb n/m	IV nm	Va $\bar{n}m$	Vb $\frac{n}{m}m$	VI n_1n_2	VII $\bar{n}_1n_2$
1			m					
2	222	$\bar{1}$	2/m	mm2		mmm		
3	32		$\bar{6}$	3m	$\bar{6}m2$		23	$m\bar{3}$
4	422	$\bar{4}$	4/m	4mm	$\bar{4}2m$	4/mmm	432	$\bar{4}3m$
6	622	$\bar{3}$	6/m	6mm	$\bar{3}m$	6/mmm		$m\bar{3}m$
							532	$m\bar{3}\bar{5}$

point groups. This means that the $8e$ position for point group $4/mmm$ (D_{4h}) is, by international convention, a position of the type (x, x, z) . We refer the reader to the tables for further information.

As a final remark, note that all the special site symmetries are subgroups of the original group $4/mmm$ (D_{4h}). This can be verified easily, using Fig. 9.12.

9.2.10.6 Crystallographic and non-crystallographic point groups Table 9.3 illustrates the relations between the crystallographic point groups and the larger set of general 3-D point groups, in particular the icosahedral point groups 532 (I) and $m\bar{3}\bar{5}$ (I_h). The limiting point groups of infinite order are shown schematically on the bottom row of the table. They are based on the highly symmetric shapes of the sphere, the cone, and the cylinder. Arrows on these shapes indicate that the object is rotating; for instance, point group ∞ is represented by a rotating cone, which has a rotation axis of infinite order along the cone axis. If the cone were at rest, then there would also be mirror planes containing the rotation axis (which is the case for point group ∞mm), but the rotation motion eliminates these mirror planes. Similarly, the top and bottom planes of the cylinder rotate in opposite directions in point group $\infty 2$, so that there are an infinite number of two-fold axes normal to the cylinder axis, but no mirror planes. A single mirror plane normal to the cylinder axis results in point group ∞/m (the rotation once again prevents the presence of mirror planes containing the rotation axis). The full cylinder symmetry of ∞/mm is obtained by eliminating all rotation motions. Finally, the sphere symmetries $\infty\infty$ and $\infty\infty m$ are distinguished from each other by the fact that, in the former, every point on the sphere surface rotates around the axis connecting it to the center of the sphere.

9.2.10.7 Examples of shapes, molecules, and crystals Table 9.4 lists, for each crystallographic point group, the names of a geometric shape, a molecule, and a mineral with that symmetry. In addition, in the second column, it lists two numbers, which represent the percentage of a population of 127 000 inorganic and 156 000 organic compounds that have that particular point group symmetry. For instance, for $2/m$ (C_{2h}) the population numbers are 34.63% for inorganic compounds and 44.81% for organic compounds. This means that out of 127 000 inorganic compounds, approximately 43 980 compounds have the monoclinic $2/m$ (C_{2h}) point group symmetry. Note that the 11 point groups corresponding to the Laue classes have the highest population numbers, indicating that the majority of inorganic compounds, around 80%, are centrosymmetric. Amongst the organic compounds, about 74% are centrosymmetric. Despite the fact that the highest cubic symmetry, $m\bar{3}m$ (O_h), only accounts for about 6.7% of all the inorganic compounds, the cubic materials play a very important role in our technological world.

9.2.10.8 Definition of generator matrices We know from the multiplication table of a group that each element of the group can be written as the product of other elements. In practice, we need to know only a few elements, and the complete group can be constructed from these by matrix multiplication. The minimum symmetry operators needed to generate the complete point group are known as the *generators*. There are only 14 matrices, D , from which all possible crystal symmetries can be derived. We will represent these matrices by symbols of the form $D^{(x)}$, where (x) is a letter ranging from (a) through (n) .⁶ The 14 matrices are defined in Table 9.5.

Table 9.6 lists, for all 32 crystallographic point groups, which generator matrices are needed to create the entire group. Note that the identity matrix, $D^{(a)}$, is always an element of the group but does not appear explicitly in Table 9.6. Note also that the selection of generators is not unique. For instance, for point group 6 (C_6), we could select just one generator (a 60° rotation around the z -axis). Instead, the table only lists generators selected from the 14 matrices in Table 9.5; the matrix representing the six-fold rotation is not one of the 14 listed, and we need two generators from the list of 14 to generate the entire group 6 (C_6). The main reason for defining the 14 matrices is that they are also used as generators for the 230 space groups, to be introduced in the next chapter.

As an example, consider point group $\bar{3}m$ (D_{3d}), which has generators $D^{(f)}$, $D^{(h)}$, and $D^{(n)}$. We begin by multiplying the generators with themselves until we obtain the identity matrix. For $D^{(f)}$ and $D^{(h)}$ this does not lead to a new symmetry operation, since $D^{(ff)} = D^{(hh)} = D^{(a)}$; for $D^{(n)}$ we obtain a new matrix:⁷

$$D^{(nn)} = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and also $D^{(nnn)} = D^{(a)}$. This is to be expected, since this matrix represents a three-fold rotation.

⁶ Additional generator matrices can be added to this list to describe non-crystallographic point groups.

⁷ We use the obvious notation that $D^{(x)}D^{(y)} \equiv D^{(xy)}$.

9.2 3-D crystallographic point symmetries

Table 9.4 Examples of molecules and crystals with symmetries belonging to the various point groups. The % population (taken from Table 3.2 in (Newnham, 2004)) consists of the percentage of 127000 inorganic and 156000 organic compounds (inorganic/organic) that are found to belong to each point group.

Point group	% population	Shape	Molecule	Crystal
1 (C_1)	0.67/1.24	Pedion	CHFCIBr SFCIO	—
$\bar{1}$ (C_i)	13.87/19.18	Pinacoid	$C_2H_2Cl_2Br_2$	Anorthite Turquoise Wollastinite
2 (C_2)	2.21/6.70	Sphenoid	H_2O_2 $C_2H_2Cl_2$	—
m (C_s)	1.30/1.46	Dome	NOCl	—
2/m (C_{2h})	34.63/44.81	Rhombic prism	H_2O_2 planar $C_2H_2Cl_2$ planar	Chlorite Datolite Epidote Gypsum Orpiment Realgar Talc Titanite
222 (D_2)	3.56/10.13	Rhombic disphenoid	C_2H_4	Edingtonite
mm2 (C_{2v})	3.32/3.31	Rhombic pyramid	CH_2Cl_2 H_2O H_2O_2 NO_2	Hemimorphite
mmm (D_{2h})	12.07/7.84	Rhombic dipyramid	$C_{10}H_8$ C_2H_4	Andalusite Aragonite Barite Chrysolite Enstatite Goethite Marcasite Sillimanite Stibnite Sulfur
3 (C_3)	0.36/0.32	Trigonal pyramid	$C_2H_3Cl_3$	—
$\bar{3}$ (C_{3i})	1.21/0.58	Rhombohedron Hexagonal prism	—	Dolomite Ilmenite Phenakite
32 (D_3)	0.54/0.22	Trigonal trapezohedron	C_2H_6	Cinnabar Low Quartz

Table 9.4 (cont.)

Point group	% population	Shape	Molecule	Crystal
3m (C_{3v})	0.74/0.22	Ditrigonal pyramid	NH ₃	Tourmaline
$\bar{3}m$ (D_{3d})	3.18/0.25	Ditrigonal scalenohedron	SF ₃ Cl ₃ C ₂ H ₆	Arsenic Brucite Calcite Corundum Hematite
4 (C_4)	0.19/0.25	Tetragonal pyramid	—	—
4 (S_4)	0.25/0.18	Isosceles tetrahedron	C ₁₂ H ₁₆	—
		Tetragonal disphenoid	C ₈ F ₄ Cl ₁₄	—
4/m (C_{4h})	1.17/0.67	Tetragonal dipyramid	—	Scapolite Scheelite
422 (D_4)	0.40/0.48	Tetragonal trapezohedron	—	—
4mm (C_{4v})	0.30/0.09	Ditetragonal pyramid	Co(NH ₃) ₄ ClBr	—
		Tetragonal prism	SF ₄ ClBr	—
$\bar{4}2m$ (D_{2d})	0.82/0.34	Ditetragonal scalenohedron	C ₂ H ₄ C ₃ H ₄ C ₈ H ₈	Chalcopyrite
4/mmm (D_{4h})	4.53/0.69	Ditetragonal dipyramid	SF ₄ Cl ₂	Rutile
		Ditetragonal prism	XeF ₄ [AuCl ₄] [−] [Ni(CN) ₄] ^{−2}	Zircon
6 (C_6)	0.41/0.22	Hexagonal pyramid	—	—
$\bar{6}$ (C_{3h})	0.07/0.01	Trigonal pyramid	Fe(OH) ₃	Nepheline
6/m (C_{6h})	0.82/0.17	Hexagonal dipyramid	—	Apatite
622 (D_6)	0.24/0.05	Hexagonal trapezohedron	—	High quartz
6mm (C_{6v})	0.45/0.03	Dihexagonal pyramid	(C ₆ H ₆) ₂ Cr	Wurtzite
$\bar{6}m2$ (D_{3h})	0.41/0.02	Ditrigonal dipyramid	BF ₃ C ₂ H ₆ C ₃ H ₃ PF ₅ SO ₃ [NO ₃] [−]	Benitoite
6/mmm (D_{6h})	2.82/0.05	Dihexagonal dipyramid	C ₆ H ₆	Beryl
		Hexagonal prism	—	Niccolite
23 (T)	0.44/0.09	Tetartoid	—	—
$m\bar{3}$ (T_h)	0.84/0.15	Diploid	—	Pyrite
432 (O)	0.13/0.01	Gyroid	—	—
$\bar{4}3m$ (T_d)	1.42/0.11	Hexakistetrahedron	CH ₄	Tetrahedrite Sphalerite

9.2 3-D crystallographic point symmetries

Table 9.4 (cont.)

Point group	% population	Shape	Molecule	Crystal
$m\bar{3}m$ (O_h)	6.66/0.12	Cuboctahedron	SF_6	Halite
		Cube	$[Fe(CN)_6]^{-3}$	Copper
		Tetrahexahedron	$[PF_6]^-$	Cuprite
		Truncated octahedron		Diamond
		Rhombic dodecahedron		Fluorite
		Octahedron		Galena
				Garnet
				Gold
				Silver
				Spinel

Table 9.5 Definition of the 14 fundamental crystallographic point symmetry matrices.

$D^{(a)} \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$D^{(b)} \equiv \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$D^{(c)} \equiv \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
$D^{(d)} \equiv \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$	$D^{(e)} \equiv \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$D^{(f)} \equiv \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
$D^{(g)} \equiv \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$D^{(h)} \equiv \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$D^{(i)} \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
$D^{(j)} \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$D^{(k)} \equiv \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$D^{(l)} \equiv \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
$D^{(m)} \equiv \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$D^{(n)} \equiv \begin{pmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	

Next, we compute the product of the two generator matrices $D^{(f)}$ and $D^{(n)}$: we find two new matrices (recall that matrix multiplication is, in general, not commutative, so we must multiply the matrices together in both possible orders):

$$D^{(fn)} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \text{and} \quad D^{(nf)} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Table 9.6 The generator matrices for the crystallographic point groups. The letters x in the generator columns refer to the transformation matrices $D^{(x)}$ of Table 9.5.

Point group	Generators	Point group	Generators	Point group	Generators	Point group	Generators
1 (C_1)	—	4 (C_4)	g	$\bar{3}$ (C_{3i})	h, n	6mm (C_{6v})	b, k, n
$\bar{1}$ (C_i)	h	$\bar{4}$ (S_4)	m	32 (D_3)	e, n	6m2 (D_{3h})	i, k, n
2 (C_2)	c	4/m (C_{4h})	g, h	3m (C_{3v})	k, n	6/mmm (D_{6h})	b, e, n, h
m (C_s)	j	422 (D_4)	c, g	$\bar{3}m$ (D_{3d})	f, h, n	23 (T)	c, d
2/m (C_{2h})	c, h	4mm (C_{4v})	g, j	6 (C_6)	b, n	$\bar{m}3$ (T_h)	c, d, h
222 (D_2)	b, c	$\bar{4}2m$ (D_{2d})	c, m	$\bar{6}$ (C_{3h})	i, n	432 (O)	d, g
mm2 (C_{2v})	b, j	4/mmm (D_{4h})	c, g, h	6/m (C_{6h})	b, h, n	43m (T_d)	d, m, l
mmm (D_{2h})	b, c, h	3 (C_3)	n	622 (D_6)	b, e, n	$\bar{m}3m$ (O_h)	d, g, h, l

This brings our total to six matrices, including the identity matrix: $D^{(a)}$, $D^{(f)}$, $D^{(n)}$, $D^{(nn)}$, $D^{(fn)}$, and $D^{(nf)}$. Finally, multiply all six matrices with the generator $D^{(h)}$, which leads to the following six new matrices:

$$D^{(ha)} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad D^{(hf)} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$D^{(hn)} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad D^{(hnn)} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

$$D^{(hfn)} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D^{(hnfn)} = \begin{pmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is straightforward to verify that any other product of these 12 matrices will generate one of the 12, in other words, we have generated the entire group of order 12. For instance, consider the product $D^{(hnhnn)}$: since $D^{(h)}$ is a diagonal matrix, we can change the order of the matrix multiplications to $D^{(hnhnn)} = D^{(aan)} = D^{(a)}$, which is one of the 12 group elements. Similarly, for $D^{(hnhfn)}$ we have $D^{(hnhfn)} = D^{(hnhfn)} = D^{(anan)} = D^{(nn)}$, which belongs to the 12 that we derived. The same procedure can be applied to the other 31 point groups, and is not too difficult to implement in a computer algorithm.

9.3

2-D crystallographic point symmetries

In 2-D, there are fewer possible point groups. We can derive the 2-D crystallographic point groups starting from the 3-D groups by eliminating those groups that contain operators that are inconsistent with two dimensions. For instance, a horizontal mirror plane has no meaning in 2-D; neither does an inversion operation. Mirror planes inclined to the horizontal plane must also be excluded. Instead of mirror planes, we only have *mirror lines*.

Table 9.7 2-D crystal systems and corresponding 2-D crystallographic point groups.	
Crystal system	Compatible point groups
Oblique	1 (C_1), 2 (C_2)
Rectangular	m (C_s), mm2 (C_{2v})
Square	4 (C_4), 4mm (C_{4v})
Hexagonal	3 (C_3), 3m (C_{3v}), 6 (C_6), 6mm (C_{6v})

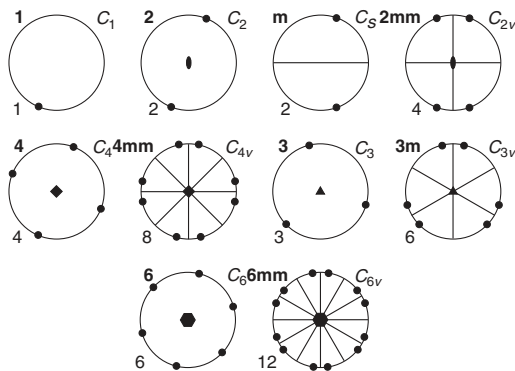


Figure 9.14 Graphical representations of the ten 2-D crystallographic point groups. Note that the general points all lie in the equatorial plane, hence they correspond to points on the projection circle.

This leaves only ten 2-D crystallographic point groups: **1** (C_1), **m** (C_s), **2** (C_2), **mm2** (C_{2v}), **3** (C_3), **3m** (C_{3v}), **4** (C_4), **4mm** (C_{4v}), **6** (C_6), and **6mm** (C_{6v}). It is customary to use the same point group symbols in 2-D and 3-D; it will be clear from the context which of the groups one refers to. Table 9.7 shows the four 2-D crystal systems, along with the point groups that belong to each system. Graphical representations of the groups are shown in Fig. 9.14.

9.4

*Magnetic point groups

9.4.1

Derivation

The derivation of the magnetic point groups (or MPGs) is relatively straightforward once one realizes that the only extra symmetry element that needs to be considered is the time-reversal operator $1'$ (E'). Therefore, we can simply start from the 32 crystallographic point

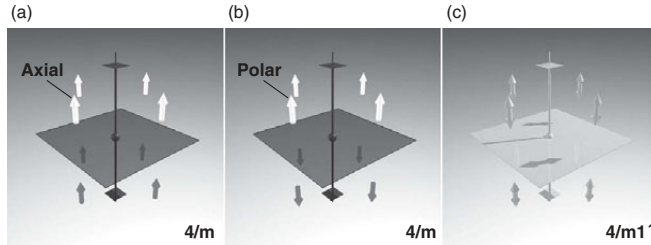


Figure 9.15 Graphical representation of the action of point group $4/m$ (C_{4h}) on (a) an axial vector and (b) a polar vector. In (c), the gray point group $4/m1'$ is depicted; at each location, two opposite magnetic moments are present.

groups (or CPGs), add the time-reversal operator to each of them, and create a new point group with twice as many elements as the original CPG. It is easy to understand why the number of symmetry elements (or the order) of the group doubles when we add one extra element: the combination of the new element with each of the original elements must also be a symmetry operator that belongs to the group. As an example, consider the tetragonal CPG $4/m$ (C_{4h}), which has the following eight elements:

$$4/m = \{1, 4, 2, 4^3, m, \bar{4}, i, \bar{4}^3\}; \quad (9.17)$$

this can be verified readily by comparison with Fig. 9.6. When we add $1'$ (E'), and combine it with all of the elements above, we find the following point group of order 16:

$$4/m1' = \{1, 4, 2, 4^3, m, \bar{4}, i, \bar{4}^3, 1', 4', 2', 4'^3, m', \bar{4}', i', \bar{4}'^3\}. \quad (9.18)$$

Note that the Hermann–Mauguin symbol for this group is obtained by taking the original CPG symbol and appending $1'$. When we repeat this with the other 31 CPGs, we obtain 32 new point groups, each with a $1'$ at the end of the original point group symbol.

Interestingly, these new point groups are known as the *non-magnetic* or *gray* point groups! To understand why this is so, we must first take a closer look at the action of a CPG on an axial vector μ . We will continue to work with $4/m$ (C_{4h}) as an example, but the following can be done for all CPGs. In the previous chapter, we have learned how the regular symmetry operators act on an axial vector. Rotations, which are operators of the first kind, treat an axial vector in the same way as a polar vector; however, operators of the second kind (mirrors and the inversion) produce different (i.e., opposite) results for polar and axial vectors. If we consider a magnetic moment vector μ oriented parallel to the four-fold axis of $4/m$ (C_{4h}), as shown in Fig. 9.15(a), then we find that all eight vectors point in the same direction; in other words, we have a net magnetic moment, which means that the point group $4/m$ (C_{4h}) is a magnetic point group. If we compare this with the action of $4/m$ (C_{4h}) on a polar vector, shown in Fig. 9.15(b), we find that there can be no net polar moment in this point group.⁸ From a materials property point of view this means that a

⁸ Of course, we would have to verify this for a random orientation of the polar vector, not just the special orientation shown in Fig. 9.15.

9.4 Magnetic point groups

crystal structure with point group $4/\mathbf{m}$ (C_{4h}) can be ferromagnetic but not ferroelectric. If we add time reversal to this point group, to obtain $4/\mathbf{m}\mathbf{1}'$, then we must, for each axial vector μ at location $\mathbf{r}$, add a second vector $\mathbf{1}'\mu = -\mu$ at the same location $\mathbf{r}$. In other words, at each equivalent location, there are two opposite moment vectors, as shown in Fig. 9.15(c), so that the net moment vanishes everywhere.⁹ Hence, this point group is known as a non-magnetic point group or a gray point group (because the mixture of black and white is gray). Since we can repeat this exercise for all CPGs, we conclude that the MPGs created by adding the time-reversal operator to the CPGs are all non-magnetic or gray. The total number of MPGs uncovered so far is therefore 64 (32 CPGs + 32 gray groups).

The next step in the derivation of the MPGs requires that we look for subgroups of the gray groups that have equal numbers of regular and time-reversing operators, but not the operator $\mathbf{1}'$ (E'). We will look for subgroups of index 2, i.e., subgroups with half as many elements as the parent gray group. Why half? Well, when we combine each of these subgroups with the time-reversal operator $\mathbf{1}'$ (E'), which doubles the number of elements in the group, we must recover the parent group. As an example, consider the gray group $4/\mathbf{m}\mathbf{1}'$; we can find three subgroups of index 2 (i.e., with eight elements), namely:

$$4'/\mathbf{m}' = \{1, 4', 2, 4'^3, m', \bar{4}, i', \bar{4}^3\}, \quad (9.19a)$$

$$4'/\mathbf{m} = \{1, 4', 2, 4'^3, m, \bar{4}, i, \bar{4}^3\}, \quad (9.19b)$$

$$4/\mathbf{m}' = \{1, 4, 2, 4^3, m', \bar{4}', i', \bar{4}'^3\}. \quad (9.19c)$$

It is easily verified that (1) each of these sets is a group, (2) each of the subgroups has an equal number of regular and time-reversing operators, and (3) when combined with $\mathbf{1}'$ (E'), each of these subgroups produces the parent group $4/\mathbf{m}\mathbf{1}'$. The notation of the point groups is also rather straightforward: we use the regular Hermann–Mauguin symbol, but add primes to those operators in the symbol that are present in their time-reversing form. For the first subgroup above, both the four-fold rotation and the mirror plane are time-reversing operations, so we add a prime to each operator. When we repeat the exercise of enumerating all the subgroups of index two for each of the 32 gray point groups, we end up with an additional 58 point groups. Along with the 64 that we uncovered before, this brings the total number of MPGs to 122.

We should note here that the terminology “122 magnetic point groups” is a bit confusing, since 32 of them are non-magnetic. It would be better to call these groups the 122 *two-color* groups, or the 122 *black-and-white* groups; then, having 32 of them with only one color (gray) makes more sense. For historical reasons, however, both terminologies are in use in the crystallographic literature. The two-color groups were first derived by Alexey Shubnikov in 1951 (Shubnikov, 1951) and are also known as the *Shubnikov point groups*.

9.4.2

Visualization of the magnetic point groups

Visualization of these new point groups is easiest using the 3-D renderings that we have been using throughout this and the previous chapters. It is possible to employ stereographic

⁹ Graphically, whenever two objects (both symmetry elements and axial vectors) with opposite temporal parity coincide in space, they are represented as gray objects.

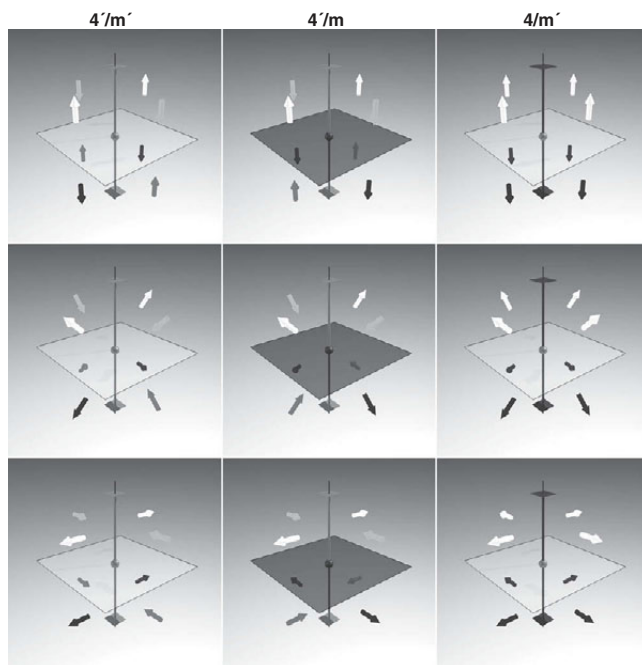


Figure 9.16 Graphical representation of the magnetic point groups $4'/m'$, $4'/m$, and $4/m'$ derived from the gray group $4/m1'$. Gray levels for the individual vectors are explained in the text.

projections to do this, but the resulting projections are very complicated and not particularly illuminating (see, for example, Joshua (1974)) because of the large number of symbols that is needed. The magnetic point groups $4'/m'$, $4'/m$, and $4/m'$ are visualized in Fig. 9.16 for three different orientations of the initial axial vector. Color versions of these figures, as well as all the other two-color point groups, can be found on the book's website.

To distinguish between regular and time-reversing operators (sometimes called *anti-symmetry operators*) we display the two types with different gray levels; regular operators are dark gray (blue in the color diagrams), time-reversing operators are light gray (yellow in the color diagrams). Furthermore, it is useful to distinguish between four different axial vector types, depending on how the vectors are related to the initial vector. If we consider the vector labeled “axial” in Fig. 9.15(a) to be the initial vector from which all the others are derived, then we can assign two parity numbers to this vector, one for the spatial parity (left-handed or right-handed) and one for the temporal parity (time-reversed or not). If we represent these two numbers as a pair of $+1$ and -1 integers, then there are four possibilities: $(+1, +1)$, $(-1, +1)$, $(+1, -1)$, and $(-1, -1)$. The operator 4 (C_4) in Fig. 9.15(a) produces three equivalent vectors at 90° from each other, but this operator does not change

9.4 Magnetic point groups

either of the parity numbers, so we draw those vectors in the same color as the initial vector. The mirror plane m (σ) changes the parity numbers from $(+1, +1)$ for the initial vector to $(-1, +1)$ for the image vector, so we represent this vector in medium gray, to indicate that it has different parity numbers. We will need two additional gray levels (colors) for the other parity pairs. The following list defines the colors used for the diagrams on the website (as an RGB triplet $[R, G, B]$ along with the official color name), as well as the corresponding gray level used for the illustrations in this book:

$(+1, +1) \rightarrow [1, 1, 1]$	white	white;
$(+1, -1) \rightarrow [0, 1, 1]$	cyan	light gray;
$(-1, +1) \rightarrow [1, 0, 1]$	magenta	medium gray;
$(-1, -1) \rightarrow [0, 0, 1]$	blue	dark gray.

We can use this color legend to determine how to color the graphical representations of the point groups $4'/m'$, $4'/m$, and $4/m'$; simply replace each symmetry operator in Eqs. 9.19 by the color corresponding to its parity. This results in the following color sets (in the same order as in Eqs. 9.19, using w , l , m , and d for white, light, medium, and dark gray, respectively):

$$4'/m' = \{w, l, w, l, d, m, d, m\}, \quad (9.20a)$$

$$4'/m = \{w, l, w, l, m, d, m, d\}, \quad (9.20b)$$

$$4/m' = \{w, w, w, w, d, d, d, d\}. \quad (9.20c)$$

Figure 9.16 shows the arrangements of axial vectors for these three point groups, and for three different orientations of the initial vector. Note that the net moment vanishes for all three of these point groups, for all initial moment orientations, so that none of these groups can describe ferromagnetic ordering. As we have seen before, the original CPG, $4/m$ (C_{4h}), does allow for ferromagnetic ordering. Table 9.8 lists all 122 magnetic or two-color point groups that are compatible with the Bravais lattices. Of those groups, 31 allow for a ferromagnetic state, i.e., a net magnetic moment; they are highlighted in bold face in the table. Note that none of the cubic groups supports ferromagnetism; this means that iron, which has a body-centered cubic crystal structure (from the point of view of the atom positions), is really not cubic when the magnetic moments are taken into account!

9.4.3

Color, charge, and time reversal

To conclude this section on the magnetic point groups, it is useful to describe briefly how these two-color or black-and-white groups can be interpreted in three different ways: to indicate color reversal, to indicate charge reversal, and to indicate time reversal (De Graef, 2010). These three interpretations are shown in Fig. 9.17. In the first interpretation, the time-reversal operator $1'$ is replaced by a *color-reversal operator* 1^c . Figure 9.17(a) shows how a two-color object (a black Greek letter Γ on a white background) can be mapped onto four different images, using (from top to bottom) the identity operator 1 , the color-reversal operator 1^c , a (vertical) mirror plane m , and a color-reversing mirror plane m^c . The pair of parities of each operation (spatial, color) is shown below each image. If we replace the

Table 9.8 List of the 122 magnetic or two-color point groups in Hermann–Mauguin notation. Bold group symbols indicate the 31 groups that allow for a ferromagnetic state.

1	11'						
$\bar{1}$	$\bar{1}1'$	$\bar{1}'$					
2	21'	2'					
m	$m1'$	m'					
2/m	$2/m1'$	2'/m'	$2/m'$	$2'/m$			
222	2221'	2'2'2					
mm2	$mm21'$	m'm'2	2'm'm				
mmm	$mmm1'$	mm'm'	$m'm'm'$	mmm'			
4	41'	4'					
$\bar{4}$	$\bar{4}1'$	$\bar{4}'$					
4/m	$4/m1'$	$4'/m$	$4/m'$	$4'/m'$			
422	4221'	$4'22'$	42'2'				
4mm	$4mm1'$	$4'mm'$	4m'm'				
42m	$42m1'$	$4'2m'$	$4'm2'$	42'm'			
4/mmm	$4/mmm1'$	$4'/mmm'$	4/nmm'm'	$4/m'm'm'$	$4/m'mm$	$4'/m'm'm$	
3	31'						
$\bar{3}$	$\bar{3}1'$	$\bar{3}'$					
32	321'	32'					
3m	$3m1'$	3m'					
$\bar{3}m$	$\bar{3}m1'$	3m'	$\bar{3}'m'$	$\bar{3}'m$			
6	61'	6'					
$\bar{6}$	$\bar{6}1'$	$\bar{6}'$					
6/m	$6/m1'$	$6'/m'$	$6/m'$	$6'/m$			
622	6221'	$6'22'$	62'2'				
6mm	$6mm1'$	$6'mm'$	6m'm'				
$\bar{6}m2$	$\bar{6}m21'$	$\bar{6}'2m'$	$\bar{6}'m2'$	$\bar{6}m'2'$			
6/mmm	$6/mmm1'$	$6'/m'mm'$	6/nmm'm'	$6/m'm'm'$	$6/m'mm$	$6'/mmmm'$	
23	231'						
$m\bar{3}$	$m\bar{3}1'$	$m'\bar{3}'$					
432	4321'	$4'32'$					
$\bar{4}3m$	$\bar{4}3m1'$	$\bar{4}'3m'$					
$m\bar{3}m$	$m\bar{3}m1'$	$m\bar{3}m'$	$m'\bar{3}'m'$	$m'\bar{3}'m$			

color-reversal operator with a *charge-reversal operator*, 1^q , then we obtain the four cases shown in Fig. 9.17(b); note that the vectors merely indicate the direction from negative to positive charge. The charge-reversing mirror is labeled by the symbol m^q . The last column, Fig. 9.17(c), shows the time-reversal case. Note that the orientation of the image moment vector for the mirror plane m in Fig. 9.17(c) is the same as the orientation of the image vector for the charge-reversing mirror plane m^q in Fig. 9.17(b). In other words, from the point of view of the orientation of the image vector, the action of a charge-reversing mirror plane on an electrostatic dipole moment vector is equivalent to the action of a regular mirror plane on a magnetic moment vector.

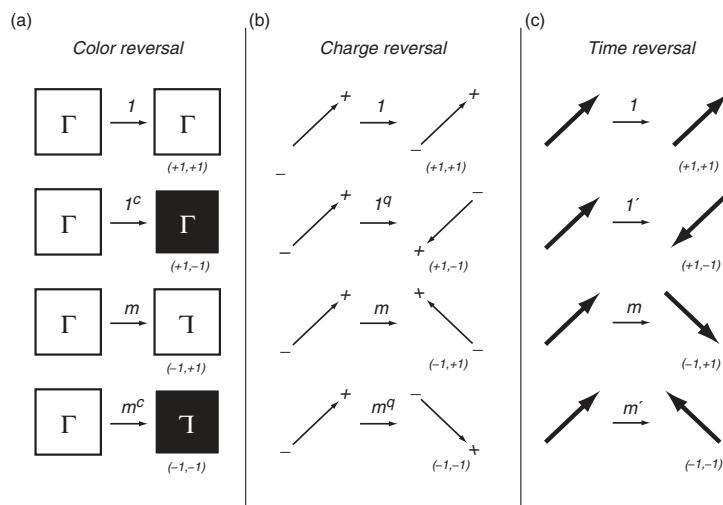


Figure 9.17 Illustration of the three possible ways of interpreting the two-color point groups: (a) black-white symmetry, (b) charge-reversal symmetry, and (c) time-reversal symmetry. The two parities involved in each of the operations are shown aside each drawing.

This has an important consequence for the relation between charge-reversal and time-reversal point groups: to obtain the equivalent symmetry group (with identical vector orientations and senses) for electrostatic dipoles, given a magnetic point group, one must replace all the time-reversed (primed) operators with odd spatial parity (i.e., mirrors and inversions) by regular odd-parity operators and vice versa. For instance, the electrostatic point group corresponding to the MPG $4'/m'$ is given by $4^q/m$, where 4^q is a charge-reversing four-fold axis. Similarly, for $4'/m$ the electrostatic equivalent becomes $4^q/m^q$. Care must be taken when multiple mirror planes are present in the MPG symbol; the electrostatic equivalent for the magnetic point group $4/mm'm'$ is $4/m^qmm$; in other words, there is a one-to-one mapping between the magnetic point groups and the charge-reversal point groups. The distinction between the electrostatic and magnetic versions of the two-color point groups becomes particularly relevant in the context of the *magnetoelectric effect*, which has received considerable interest in recent years (Fiebig, 2005). In the presence of both electric and magnetic fields, neither the time-reversal operator, $1'$, nor the spatial inversion, i , can be symmetry operations, which restricts the presence of the magnetoelectric effect to only 58 of the 122 two-color point groups.

In this chapter we have introduced the concept of a point group, a closed set of associative symmetry operators that includes a unit element, and contains the inverse of each

operator. Then we derived the 32 crystallographic point groups by enumerating all the different combinations of point symmetry operators that are consistent with the Bravais translational symmetry. We discussed point group classes, descent in symmetry (group-subgroup relations), chirality and enantiomorphism, matrix representations, special positions and orbits, and generator matrices, as well as molecular and solid examples of each point group. We concluded the chapter with a more advanced section on the derivation and visualization of the 122 magnetic (two-color, black-and-white, Shubnikov) point groups.

9.6

Historical notes

Group theory, invented by the gifted French mathematician **Evariste Galois** (1811–32), occupies an important position amongst the mathematical theories. Galois lived during a tumultuous period in French history. He was born at the peak of Napoleon's power in 1811, just a few years before his historic loss at Waterloo. As a teenager, he became interested in mathematics, in particular in the theory of equations, and he published his first paper on continued fractions in 1829, at age 17! Mathematicians at that time were very interested in the solubility of polynomial equations. While the solutions to the quadratic equation had been known to the Babylonians and the Greeks, the general cubic and quartic equations had been solved only recently (see [Livio \(2005\)](#) for an entertaining account), and the next equation, the quintic, appeared to resist all attempts at finding a mathematical expression for its solutions. Galois developed what is now known as the *Galois theory*, a precursor of modern group theory, which uses symmetry properties of the polynomial equations to decide whether or not the solutions can be written down using simple functions. He showed that the solution of the quintic equation cannot be written down using only additions, subtractions, multiplications, divisions, and roots.

Despite his young age, Galois was one of the most influential mathematicians of the nineteenth century. His scientific work includes results on elliptic functions and Abelian integrals. He died at age 20 from wounds received during a pistol duel ([Livio, 2005](#)). After his death, his brother, Alfred, and a friend collected all of his writings and, eventually, another influential French mathematician, Joseph Liouville, published them in 1846. From then on, group theory became an important mathematical field, with numerous applications in symmetry, physics, biology, language, music, and so on.

The contributions of **J. F. C. Hessel** ([Fig. 9.18\(b\)](#)) were discussed in the historical section of Chapter 3, on page 72.

9.7

Selected problems

- (i) *Plane group*: Consider the symmetric structure shown in [Fig. 9.19](#).
 - (a) Determine the unit cell (unit mesh).
 - (b) Show all symmetry elements in the cell.
 - (c) Determine the 2-D plane group for this structure.

9.7 Selected problems

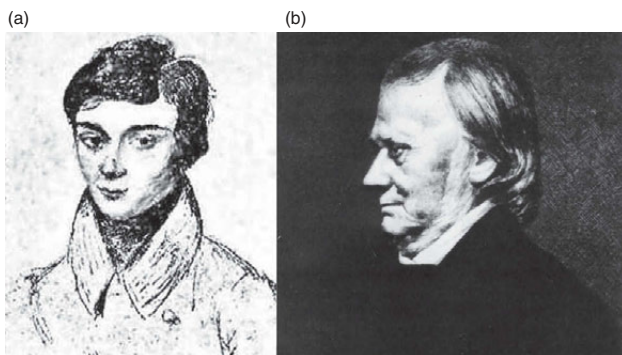


Figure 9.18 (a) Evariste Galois (1811–32) and (b) J. F. C. Hessel (1796–1872) (picture courtesy of J. Lima-de-Faria).

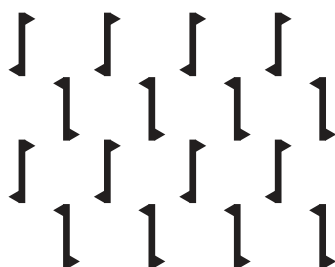


Figure 9.19 Symmetric 2-D structure.

- (ii) *Point group*: Consider the dihedral point group, $\bar{6}m2$ (D_{3h}) constructed from two generators, $\tilde{C}_6^z$ and 2^x , i.e., a six-fold roto-inversion operation about the z -axis and a two-fold rotation about the x -axis.
 - (a) Express 3-D matrix representations for the two generators of the group in the hexagonal basis.
 - (b) By repeated operation with these two matrices, determine matrix representations for all elements of the group. What is the order of this group?
 - (c) Draw a stereographic projection for this group.
 - (d) Determine all the entries in this group's *multiplication table* using either a stereographic projection or direct matrix multiplication.
- (iii) *Subgroups*: Determine the operations that are lost in the reduction of symmetry from $6/mmm$ (D_{6h}) to 622 (D_6). Show this graphically by means of a stereographic projection.
- (iv) *Magnetic point groups*: By analogy with the derivation of the magnetic point groups related to the CPG $4/m$ (C_{4h}) in Section 9.4, derive all the magnetic points groups that are related to the CPG $2/m$ (C_{2h}).